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Reflections on Data Science 2025

Journal

Authors: Theodoros V. Giannouchos et al., Calling BS on "Characteristics and risk factors for COVID-19 diagnosis and adverse outcomes in Mexico: an analysis of 89,756 laboratory–confirmed COVID-19 cases" [1]

The authors claim that smokers are less likely to be diagnosed with COVID-19 than non-smokers. This conclusion stems from serious methodological flaws that led to unreliable and potentially dangerous findings.

The dataset used included only laboratory-confirmed COVID-19 cases in Mexico, excluding symptomatic but untested individuals. This introduces censoring bias, which may underrepresent smokers who avoid or delay medical care. Such selective exclusion results in a statistical trap, where conclusions are drawn from a non-representative sample.

Another flaw lies in the self-reported smoking status, which allows under-reporting. The study also fails to consider former smokers, who may still be at elevated risk but are grouped as non-smokers, further skewing results.

Moreover, confounding factors such as socioeconomic status, occupation, and healthcare access, variables correlated with both smoking and COVID-19 risk, are not properly accounted for.

We applied several tools from the course to identify the statistical traps in the study. Specifically, we used the concept of censoring to understand how excluding untested individuals biases the sample toward those more likely to seek testing. We also applied selection bias, noting that smokers may be underrepresented in the dataset if they are less likely to seek medical attention due to social stigma [2]. Finally, we used misclassification analysis to highlight the flawed categorization of current vs. former smokers. These tools helped us diagnose why the study's conclusions about smoking and COVID-19 risk are unsupported and misleading.

The paper was later retracted due to a conflict of interest: a lead author was affiliated with an NGO funded by the tobacco industry.

This kind of misinformation is not only academically flawed, but it poses severe public health risks. Claims that smoking reduces COVID-19 risk can be weaponized by tobacco companies, undermine cessation efforts, and contribute to harmful behaviors among vulnerable populations.

References

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Reflections on Data Science 2025

Journal

Authors: Mishra, T., Wang, M., Metwally, A.A. et al. Calling BS on "Pre-symptomatic detection of COVID-19 from smartwatch data." [1]

The authors claim that smartwatches like the Apple Watch and Fitbit can detect COVID-19 infections before symptom onset, thanks to subtle physiological changes in heart rate variability and resting pulse. The conclusion: Wearable technology may become a powerful early warning system in future pandemics.

Although the idea is compelling and great for press releases, it is based on a foundation of unfair comparisons and flawed benchmarking that undermine the validity of the findings.

The study compared physiological data from infected individuals with their own baseline data from 'healthy' periods, but did not control for other infections, stress or changes in activity level. By not comparing COVID-positive individuals with a COVID-negative control group matched properly, the authors created a classic case of apples to oranges. Any deviation from baseline could look meaningful if you never test whether similar deviations occur in uninfected individuals.

The study also lacks alternative hypotheses: Could changes in heart rate and sleep data be caused by anxiety about the pandemic? Increased screen time? Poor sleep from doom-scrolling? The researchers never test these explanations, focusing solely on COVID as the cause.

Using tools from the course, this was identified as a case of unfair comparison, where the "treatment" and "control" conditions are not equivalent. We also used counterfactual reasoning: What would we expect the wearable data to look like if the person was not infected? That question was completely ignored in the paper.

The study's bold claims contributed to widespread public misunderstanding. By suggesting that wearable devices alone can detect COVID-19 before symptoms appear, it risked creating unrealistic expectations about their diagnostic capabilities. This may lead consumers to place undue trust in personal health data, while public health officials caution that such misinterpretations could encourage self-diagnosis and delay proper medical care.

This type of BS, dressed up in machine learning jargon and backed by tech company funding, illustrates how data-driven claims can mislead when not grounded in fair comparisons or robust statistical reasoning.

References

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